

An emoji-aware multitask framework for multimodal sarcasm detection

Dushyant Singh Chauhan^{a,*}, Gopendra Vikram Singh^{a,1}, Aseem Arora^a, Asif Ekbal^a, Pushpak Bhattacharyya^b

^a Department of Computer Science and Engineering, Indian Institute of Technology Patna, India

^b Department of Computer Science and Engineering, Indian Institute of Technology Bombay, India

ARTICLE INFO

Article history:

Received 23 May 2022

Received in revised form 15 September 2022

Accepted 16 September 2022

Available online 26 September 2022

Keywords:

Multimodal sarcasm
Multimodal sentiment
Multimodal emotion
Emoji
MUSTARD dataset
Deep learning

ABSTRACT

Sarcasm is a case of implicit emotion and needs additional information like context and multimodality for better detection. But sometimes, this additional information also fails to help in sarcasm detection. For example, the utterance “Oh yes, you’ve been so helpful. Thank you so much for all your help”, said in a polite tone with a smiling face, can be understood easily as non-sarcastic because of its positive sentiment. But, if the above message is accompanied by a frustrated emoji 🙄, the negative sentiment of the emoji becomes evident, and the intended sarcasm can be easily understood. Thus, in this paper, we propose the *SEEmoji MUSTARD*, an extension of the multimodal MUSTARD dataset. We annotate each utterance with relevant emoji, emoji’s sentiment, and emoji’s emotion. We propose an emoji-aware-multimodal multitask deep learning framework for sarcasm detection (i.e., primary task) and sentiment and emotion detection (i.e., secondary task) in a multimodal conversational scenario. Experimental results on the *SEEmoji MUSTARD* show the efficacy of our proposed emoji-aware-multimodal approach for sarcasm detection over the existing models.

© 2022 Elsevier B.V. All rights reserved.

1. Introduction

Usage of comments that mean the reverse of what is said and made to injure the feelings of others or to attack something humorously; with this definition, we can say that sarcasm is very important in our daily lives. But sarcasm is not all about hurting someone’s feelings; it also adds some laughter to the vocabulary used in our daily communication. Sarcasm is always implicit; for example: “What a beautiful essay you’ve written,” he said with heavy sarcasm as s/he looked at the empty notebook. In this example, the explicit sentiment is positive, but the implicit sentiment is negative, which makes this example sarcastic.

When we know that sarcasm is implicit, we can also agree that sometimes just going through the utterance text is not enough to understand sarcasm. For example, the utterance (only text) “It’s just a privilege to watch your mind at work” is positive in nature. If it is intended sarcastically, it is next to impossible to understand. Suppose this utterance is multimodal and is accompanied by a video of the facial expressions and tone of the speaker. In that

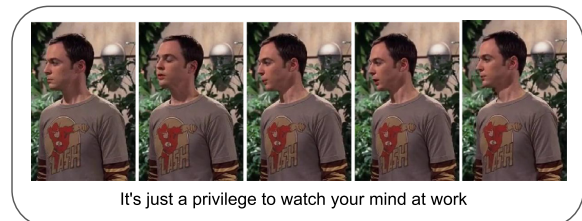


Fig. 1. Examples to show the importance of speaker’s sentiment and emotion for sarcasm detection.

case, it can be easily understood that the utterance is sarcastic [1]. We show some visual expressions in Fig. 1, and it is quite visible that the person is very frustrated with his work. This additional information (i.e., acoustic and visual) help to understand that the utterance is sarcastic.

Emojis are a trending topic because they provide an expressive way to convey sentiment and emotion. They are also a convenient way of understanding the implicit sentiment and emotion of the utterance. As sarcasm is closely related to the understanding of implicit sentiment and emotion, we can hypothesize that emojis should help to understand if there is any intended sarcasm in the utterance or not. Sometimes the implicit sentiment and emotion are not so evident in the utterance. In such utterances, emojis

* Corresponding author.

E-mail addresses: 1821cs17@iitp.ac.in (D.S. Chauhan), gopendra_1921cs15@iitp.ac.in (G.V. Singh), aseem_1911mc02@iitp.ac.in (A. Arora), asif@iitp.ac.in (A. Ekbal), pb@cse.iitb.ac.in (P. Bhattacharyya).

¹ The first two authors contributed equally to this work and are jointly the first authors.

play a crucial role in understanding sarcasm. For example, if the same utterance, "It's just a privilege to watch your mind at work," is said with neutral facial expressions and a polite tone. It becomes very hard to understand the intended sarcasm, even with multimodality. But, if an emoji is also present in this utterance, "It's just a privilege to watch your mind at work 🤔", then the intended sarcasm becomes pretty evident.

Even though sarcasm is related to emotion and sentiment, sarcasm detection is very challenging, so everyone treats this task separately. But if we introduce emojis, somewhat sarcasm becomes easy to compare before. The main contributions and attributes of our proposed research are as follows:

1. We propose the *SEEmoji MUsTARD*, an extension of the multimodal MUsTARD dataset [1]. We manually annotate each utterance with relevant emoji, emoji's sentiment, and emoji's emotion;
2. We propose an emoji-aware-multitask learning framework for multimodal sarcasm detection. In our multitask framework, we treat sarcasm detection as the primary task, whereas emotion and sentiment analysis are auxiliary tasks;
3. We propose a Gated Emoji-aware-multimodal Attention mechanism for sarcasm detection;
4. We present state-of-the-art systems for sarcasm detection in the multimodal scenario.

The rest of the paper is organized as follows. Section 2 discusses the work related to multimodal sarcasm, sentiment, emotion, and emoji. Section 3 describes the notations used in this paper, followed by Section 4, describes input features. In Section 5, we describe and explain our proposed methodology in detail. Next, we discuss our dataset with annotation guidelines in Section 6. In Section 7, we describe experiments and necessary analysis. Finally, the paper is concluded in Section 8 with directions for future research and ethical declaration in Section 9.

2. Related work

In this section, we present a brief survey on the research for multimodal sarcasm, sentiment, emotion, and emoji analysis.

2.1. Sarcasm

The sheer subtlety with which sarcastic comments could be made, makes it challenging to understand them. This makes sarcasm detection an interesting research problem, and a review of the existing research [2–5] also indicates the same. Multimodal data contains more information than text and should provide additional evidence for sarcasm detection. Razali et al. [6] compared different papers and showed that all the papers have the same conclusion that multimodal is good for sarcasm than "text". Similarly, Das et al. [7] argued that word-level information limited the system performance and suggested that the multimodal approach is better for sarcasm detection than the unimodal approach.

Nowadays, social media, like Twitter, Instagram, etc., are good platforms for sarcasm or making fun of others. Thus, Schifanella et al. [8] collected data from social media and concatenated textual and visual data for input. The author proposed two frameworks for sarcasm detection, where the first framework tries to exploit visual semantics and concatenate for the second approach. They used pre-trained ImageNet weights to detect multimodal sarcasm detection. Similarly, [9] fused all three modalities, i.e., text feature, image feature, and image attributes, with a hierarchical fusion model for detecting sarcasm. Whereas a 2D-Intra-Attention [10] is proposed to find the relationship between image and words based image-text model for sarcasm detection.

The authors used BERT embedding for text and ResNet for image feature extraction.

Castro et al. [11] introduced a new multimodal sarcasm MUsTARD dataset with all three modalities, i.e., text, acoustic, and visual, and showed that multimodal data helps better sarcasm detection than unimodal data. Whereas Chauhan et al. [1] used this MUsTARD dataset and manually annotated the MUsTARD dataset with implicit/explicit sentiment and implicit/explicit emotions because sarcasm is implicit, so the relation between the implicit and explicit sentiment/emotion help to predict sarcasm. The author also proposed two attention mechanisms, Inter-segment Inter-modal Attention (I_e -Attention) and Intra-segment Inter-modal Attention (I_a -Attention), for detecting Sarcasm, Emotion, and Sentiment in a multitask framework.

In another work, Attardo et al. [12] concluded markers for irony and sarcasm by studying multimodal stimuli. The authors collected data from television comedy show. Whereas Chauhan et al. [13] proposed two attention modules, i.e., Inter-task Relationship Module (iTRM) and Inter-class Relationship Module (iCRM), for detecting humor, sarcasm, offensive, motivational, sentiment, for this they use Memotion dataset. The objective of this paper was to establish the relationships among multiple tasks. Similarly, Pan et al. [14] proposed a BERT-based model concentrating on intra and inter-modal incongruity for multimodal sarcasm detection.

2.2. Sentiment

Sentiment analysis [15–19] refers to detecting the polarity (i.e. positive, negative, or neutral) from given input. Knowing the sentiment could help make a better decision in tasks that involve understanding the scenario and responding accordingly. Sentiment recognition is one of the challenging tasks, but somehow multimodality makes it a bit easy. These days people are more reliant on video messages to express their sentiments.

Kaur et al. [20] compared different multimodal sentiment modal papers and showed why and how multimodality plays a crucial role in detecting sentiment analysis in different areas. Similarly, a survey [17] on sentiment defined the problems during multimodality sentiment analysis and reviewed the development in different modalities of spoken reviews, images, video blogs, human-machine, human-human interactions, and sentiment analysis.

Ghosal et al. [15] proposed a recurrent neural network-based multimodal attention framework for multimodal sentiment analysis. An auto-encoder-based approach to learn the modalities interaction for sentiment and emotion analysis is proposed in [18]. The author performed experiments on five standard multimodal affect analysis datasets (YouTube, MOUD, ICT-MMMO, CMU-MOSI, and CMU-MOSEI).

In another work, a multimodal fusion model that exclusively used visual and acoustic features to analyze spoken sentences for sentiment analysis is proposed in [21]. They used the CMU-MOSEI multimodal dataset. Whereas Perez et al. [22] showed the importance of multimodality in finding the sentiment and proposed that joint use of visual, acoustic, and linguistic modalities can lead to error rate reductions.

2.3. Emotion

Emotion detection [23–25], like sentiment analysis, is an important and challenging task and can help in unveiling information that could be used for different purposes, such as the behavior of a person, response generation for conversational agents, etc. Sometimes text itself is not enough to find the emotion, or in

general words, we can say that unimodality sometimes fails to detect the emotion.

Dai et al. [26] talked about unseen emotions and the problem of detecting them and suggested a modality-transferable model with emotion embeddings to tackle the above problems. A Transformer-based joint-encoding (TBJE) model is proposed for detecting the multimodal sentiment and emotion in [27]. In another work, Choi et al. [28] proposed Convolution Attention Network to find the hidden representations between just speech and text data. They used the proposed method on the CMU-MOSEI dataset to detect multimodal emotion.

A new multimodal dataset named MELD is proposed to show the importance of multimodality and context for detecting emotion in conversation in [23]. Similarly, Hazarika et al. [29] proposed an interactive conversational memory network, a model which extracts multimodal features from conversational videos and showed that emotion detection outperforms results for multiple classifications. In a survey, Sharma et al. [30] showed that emotions are more talkative than words during a conversation. Also, he showed how multimodality helps the conversation to find emotion.

Emotion analysis is a challenging task but it also helps to find the other tasks as well like sentiment [25], emoji [31,32], humor [13], etc. Akhtar et al. [25] proposed a context-level inter-modal attention framework for detecting the sentiment and emotion of the utterance. They used their approach on the CMU-MOSEI dataset for multimodal emotion and sentiment. In another work, Shueb and de Melo [31] proposed a new dataset, Emo-Tag1200, and tried to find the connection between emotion and emojis in an unsupervised manner. Whereas, Illendula et al. [32] investigated the domain knowledge for emoji to detect the emotion that they used three modalities (i.e., text, emojis, and images) and also showed that if we combined the emoji with text encodes, better information than separately.

2.4. Emoji

Emojis are a trending topic these days. With the rampant usage of emojis, related tasks such as emoji prediction [33–35] have become essential. Therefore many researchers have done work on its usage in different areas such as for emoji recommendation in instant messages [36], emoji sense disambiguation [37], understanding crisis events [35], building emotion classifiers [38], sentiment detection [39–41] and emotional response generation [42].

Barbieri et al. [33] proposed several LSTM-based frameworks for single-label emoji prediction. Similarly, an RNN-based approach with emojis was used to study solidarity in social media. They involved temporal and geospatial diffusion of emojis. Whereas Jin et al. [34] proposed an ensemble framework based on Naive Bayes, Logistic Regression, and Random Forests for the multilingual emoji prediction task. Lately, Ma et al. [43] proposed a transformer-based network for multi-label emoji prediction. In another work, Barbieri et al. [44] proposed a multimodal fusion-based approach for multimodal emoji prediction. The authors used only text and images for emoji prediction.

Zhou et al. [42] solved the problem of generating emotional responses. They used a reinforced conditional variational encoder-based approach and trained a deep generative model. The emotion of the generated text was controlled with emojis. An embedding for Unicode emojis named emoji2vec, where emoji embedding learned from a short description of emoji, is proposed in [45].

In comparison to the existing systems, we first extend the MUS-ARD dataset [1] by manually annotating each utterance with emoji, emoji's sentiment, and emoji's emotion. Thereafter, we propose an

emoji-aware multitask framework along with Emoji-aware Multimodal Attention mechanisms to leverage the emojis for sarcasm detection. Further, to the best of our knowledge, this is the very first attempt at solving the multimodal sarcasm, sentiment, and emotion with emoji in a multitask framework. We demonstrate through a detailed empirical evaluation that sarcasm detection can be improved significantly if we successfully leverage the knowledge of emoji, emoji sentiment, and emoji emotions in an effective multitask framework.

3. Notation

For convenience, we show all the notations present in the paper in Table 1.

Table 1
Notations.

	Notation	Description
Input	u/c	Utterance/context
	E^m	Emoji
	$T_{u/c}$	Text of utterance/context
	$A_{u/c}$	Acoustic of utterance/context
	$V_{u/c}$	Visual of utterance/context
	$E^m_{u/c}$	Emoji of utterance/context
	$SP_{u/c}$	Speaker of utterance/context
Labels	I_s	Implicit sentiment
	I_e	Implicit emotion
	E_s	Explicit sentiment
	E_e	Explicit emotion
	E^m_s	Emoji's sentiment
	E^m_e	Emoji's emotion
	S^c	Sarcasm

4. Input features

Each sample in the annotated dataset consists of an utterance (u) accompanied by its context and labels (sarcasm,² implicit sentiment, implicit emotion, explicit sentiment, explicit emotion, emoji's sentiment, and emoji's emotion). The context associated with the utterance consists of multiple utterances (say, N) that precede the corresponding utterance in the dialogue. Each utterance and its' context are associated with its' speaker, i.e., speaker of utterance and speaker of context, respectively. We describe the utterance and its context for all the modalities below;

4.1. Text features

4.1.1. Utterance

Let us assume, in an utterance, there are n_t number of words $w_{1:n_t} = w_1, \dots, w_{n_t}$, where $w_j \in \mathbb{R}^{300}$. Each word, w_j , represents a vector which is obtained using *fastText* word embedding [46]. We use a bi-directional Gated Recurrent Unit [47] (*BiGRU*) to learn the contextual relationship between the words. We then extract utterance level features by taking last word embedding which is denoted by T_u .

$$T_u = \text{BiGRU}(w_{1:n_t} = w_1, \dots, w_{n_t})[-1] \quad (1)$$

If speaker information (SP_u) is available then we concatenate with T_u to get speaker-aware textual utterance which is denoted by $T_u + SP_u$.

$$T_u + SP_u = \text{Concat}(T_u, SP_u) \quad (2)$$

² The sarcasm is associated with only utterance (u) not with context utterances.

4.1.2. Context

Let's assume that there are N_c number of utterances in the context and each utterance has n_{tc} words. To obtain utterance-level representations, words of each utterance are passed through a BiGRU, and we then take the last word's embedding. We denote i_{th} utterance in the context by T_c^i .

$$T_c^i = \text{BiGRU}(w_{1:n_{tc}}^i = w_1^i, \dots, w_{n_{tc}}^i)[-1] \quad (3)$$

If speaker information is available then we concatenate speaker information (SP_c^i) with each context utterance (T_c^i). Finally, we concatenate all the speaker-aware textual utterances to get the context-level features.

$$T_c + SP_c = \text{Concat}((T_c^1 + SP_c^1), (T_c^2 + SP_c^2), \dots, (T_c^{N_c} + SP_c^{N_c})) \quad (4)$$

4.2. Visual features

4.2.1. Utterance

Given n_v number of visual frames w.r.t. an utterance, we take the average of all the frames to extract the utterance-level information. If speaker information is available then speaker information (SP_u) is concatenated with utterance (V_u). We denote this as $V_u + SP_u$, where $V_u \in \mathbb{R}^{2048}$.

4.2.2. Context

Given n_{vc} number of visual frames w.r.t. all the context utterances, we take the average of all the visual frames and use it as context-level information; V_c denotes this. We do not consider speaker information as the dataset does not contain visual frames at the utterance level.

4.3. Acoustic features

4.3.1. Utterance

Given n_a number of frames for the acoustic w.r.t. an utterance, we take the average of all the frames to extract the utterance level information. If speaker information is available then speaker information (SP_u) is concatenated with utterance (A_u). We denote this as $A_u + SP_u$, where $A_u \in \mathbb{R}^{283}$.

4.3.2. Context

Please note that as acoustics often contain information from different speakers, background noise, background laughter cues, etc., we do not consider any context information for acoustics. So, we use [] empty list for the context of acoustic. It might be challenging to disambiguate this with the laughter part of the conversation. Whereas the visual will not face the same problem as it majorly contains the speaker's image and the information about sentiment and emotion.

Please note that we take the average of the acoustic and visual features across the utterances for a fair comparison with the state-of-the-art systems [1,11].

4.4. Emoji features

4.4.1. Utterance

There is one emoji associated with each utterance. The pre-trained emoji embeddings, of dimension 300, are obtained using emoji2vec³ [45]. This is denoted by E_u^m .

4.4.2. Context

There are N_c number of utterances in the context, and one emoji is associated with each utterance. To extract context-level emoji features, we concatenate all emoji embeddings denoted by E_c^m .

5. Methodology

In this section, we describe our proposed methodology, where we aim to leverage the emoji information for solving the problem of multimodal sarcasm detection in a multitask framework. We propose a multitask deep learning framework for sarcasm detection (primary task), and sentiment and emotions detection (secondary tasks) in a multimodal conversational scenario. We depict the overall architecture in Fig. 2. The extended dataset and source code are available at <https://www.iitp.ac.in/~ai-nlp/ml/resources.html#SEEmoji-MUSTARD>.

5.1. Model description

We have already extracted the utterance features (i.e., $T_u + SP_u$ for text, $V_u + SP_u$ for visual, $A_u + SP_u$ for acoustic, and E_u^m for emoji) and context features (i.e., $T_c + SP_c$ for text, V_c for visual, and E_c^m for emoji). Afterwards, we concatenate utterance features and context features;

$$\begin{aligned} T_{cat} &= \text{Concat}([T_u + SP_u], [T_c + SP_c]) \\ V_{cat} &= \text{Concat}([V_u + SP_u], [V_c]) \\ A_{cat} &= \text{Concat}([A_u + SP_u], []) \\ E_{cat}^m &= \text{Concat}([E_u^m], [E_c^m]) \end{aligned} \quad (5)$$

where [] denotes the empty list as there are context-level features for acoustic. We then pass T_{cat} , V_{cat} , & A_{cat} through the linear layer separately (c.f. Fig. 2);

$$\begin{aligned} L_T &= \text{Linear}(T_{cat}) \\ L_V &= \text{Linear}(V_{cat}) \\ L_A &= \text{Linear}(A_{cat}) \end{aligned} \quad (6)$$

We concatenate all the modalities together and pass through another linear layer to obtain multimodal representation (MR).

$$\begin{aligned} TAV_{cat} &= \text{Concat}(L_T, L_A, L_V) \\ MR &= \text{Linear}(TAV_{cat}) \end{aligned} \quad (7)$$

Finally, we apply two Softmax layers to predict the I_s & E_s and two Sigmoid layers⁴ to predict the I_e & E_e .

$$\begin{aligned} I_s &= \text{Softmax}(MR) \\ E_s &= \text{Softmax}(MR) \\ I_e &= \text{Sigmoid}(MR) \\ E_e &= \text{Sigmoid}(MR) \end{aligned} \quad (8)$$

Similarly, we pass emoji embedding (i.e., E_{cat}^m in Eq. (5)) through a linear layer. Then, we apply two softmax layers to predict the E_s^m and E_e^m and concatenate them (c.f. Fig. 2) to obtain emoji representation (ER).

$$\begin{aligned} L_E^m &= \text{Linear}(E_{cat}^m) \\ E_s^m &= \text{Softmax}(L_E^m) \\ E_e^m &= \text{Softmax}(L_E^m) \\ ER &= \text{Concat}(E_s^m, E_e^m) \end{aligned} \quad (9)$$

The objective of ER is to enhance the information about the emoji's behavior or nature. Then, we pass these outputs, i.e., MR, I_s , E_s , I_e , E_e , and ER, to Gated Emoji-aware Multimodal Attention to learn the relationship between multimodal and emoji.

³ <https://github.com/uclnlp/emoji2vec>

⁴ There can be more than one emotion per utterance; this is why we use Sigmoid.

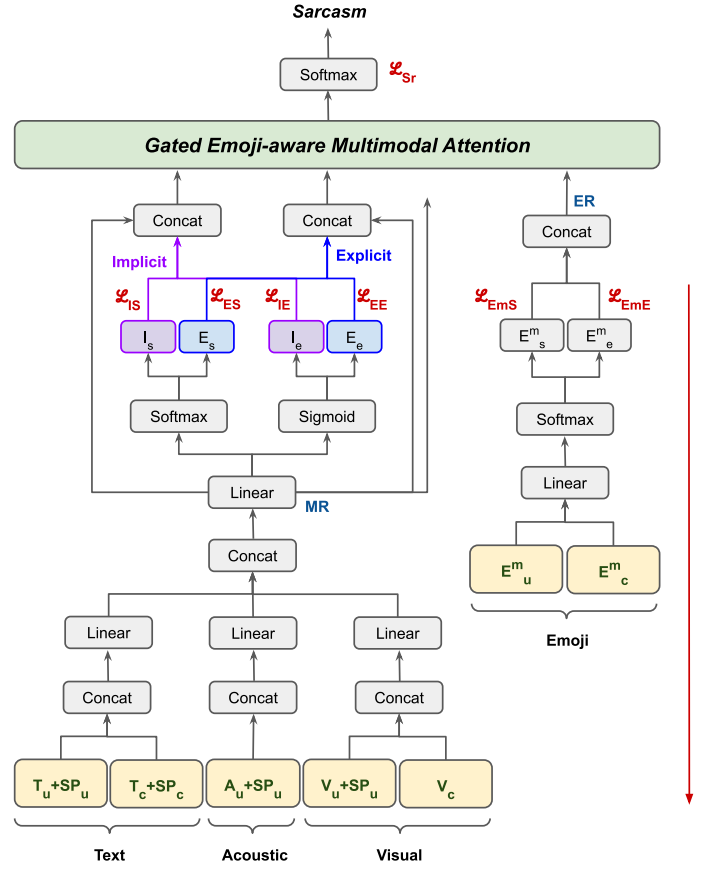


Fig. 2. Overall architecture of the proposed Emoji-aware Multimodal Sarcasm detection framework.

5.1.1. Gated emoji-aware multimodal attention

We propose a Gated Emoji-aware Multimodal Attention (GEMA) to learn the joint association between the multimodal & emoji and emphasize the contributing features by paying more attention to these. For example, if multimodal is helping, attention will put more weight on multimodal instead of emojis and vice versa. For GEMA, we obtain three different representations to capture the relationship between the multimodal inputs and emoji, as explained below:

1. Emoji-aware Implicit Multimodal Representation (EIMR):

To capture the relationship between emoji and *implicit* behavior of multimodal data, we obtain EIMR by concatenating ER with MR, I_s and I_e .

$$EIMR = \text{Concat}(ER, MR, I_s, I_e) \quad (10)$$

Those cases where model fails to predict implicit sentiment/emotion correctly⁵ which may not be helpful for sarcasm detection but with emoji representation, it can easily detect sarcasm. For example, as we have already explained that utterance “It’s just a privilege to watch your mind at work” If said with neutral facial expressions and a polite tone, it becomes very hard to understand the intended sarcasm because model cannot predict implicit sentiment as negative. But, if an emoji is also present in this utterance, “It’s just a privilege to watch your mind at work 🤔”, then the intended sarcasm becomes pretty evident.

2. Emoji-aware Explicit Multimodal Representation (EEMR):

To capture the relation between emoji and *explicit* behavior

of multimodal data, we obtain EEMR by concatenating ER representation with MR, E_s and E_e .

$$EEMR = \text{Concat}(ER, MR, E_s, E_e) \quad (11)$$

As we know sarcasm is implicit but it is also dependent on explicit because only implicit sentiment/emotion cannot detect sarcasm. For sarcasm detection we need both implicit and explicit sentiment/emotion. Generally, we believe that if implicit sentiment is negative and explicit sentiment is positive then there will be high chance to be sarcasm. But what if model fails to predict explicit sentiment as positive while predict negative then emoji 😊 with positive sentiment will help in sarcasm detection.

3. **Emoji-aware Multimodal Representation (EMR):** To capture the relation between emoji and multimodal data without including sentiment and emotion information, we obtain the EMR by concatenating MR representation with ER.

$$EMR = \text{Concat}(ER, MR) \quad (12)$$

Those cases where an utterance has same implicit/explicit sentiment then model may fail to detect sarcasm but with emoji representation, model may detect sarcasm easily. For example, “So? Do cocaine smugglers write “co-caine” on the box? 🙄” this utterance has negative implicit/explicit sentiment but positive sentiment of 🙄 helps to detect sarcasm easily.

Gated Multimodal Unit (GMU): We first employ a gated architecture (GMU) [48] to refine or filter out the noise from ER, EIMR, EEMR, and EMR. We depict overall architecture of GMU in Fig. 3. Traditionally, the GMU module is used for information fusion

⁵ Implicit sentiment and emotion detection itself a very big challenge.

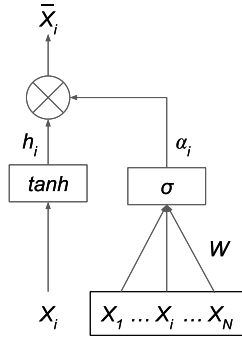


Fig. 3. Gated multimodal unit.

which finds a refined representation of an input based on the combination of all other inputs. Lets suppose, there are N inputs (say $(X_1, X_2, \dots, X_N)$) and each input representation (X_i) is passed through an activation function (i.e. $\tanh$) to encode the input representation. Further, the encoded representation is filtered through a gating mechanism ($\text{sigmoid}(\sigma)$) computed on all the available input sources. This essentially computes the significance of an input source ($\overline{X_i}$) w.r.t. other available sources of information. The equations for computing GMU for i th source (X_i) is as follows:

$$\begin{aligned} h_i &= \tanh(W_i \cdot X_i) \\ X &= \text{Concat}(X_1, \dots, X_i, \dots, X_N) \\ \alpha_i &= \sigma(W \cdot X) \\ \overline{X_i} &= h_i * \alpha_i \end{aligned} \quad (13)$$

Similarly, we compute the gated representations, i.e., $\overline{X_{ER}}$, $\overline{X_{EIMR}}$, $\overline{X_{EEMR}}$, & $\overline{X_{EMR}}$ and then we concatenate them to compute GMU output (i.e., GMU_{out}).

$$\begin{aligned} GMU_{out} &= \text{Concat}(\overline{X_{ER}}, \overline{X_{EIMR}}, \overline{X_{EEMR}}, \overline{X_{EMR}}) \\ GMU_{out} &= \text{resize}(GMU_{out}, [-1, 4, \text{size}(GMU_{out})/4]) \end{aligned} \quad (14)$$

After getting GMU_{out} , we then compute *cross-multiplication* between GMU_{out} that account for the *cross-gated* scores.

$$M = GMU_{out} \cdot GMU_{out}^T \quad (15)$$

where T denotes the transpose of the matrix. Afterwards, we compute the probability distribution scores over each multiplicative gated score of matrices M using a softmax function. This essentially computes the *cross-gated* attentive weights.

$$P(i, j) = \frac{e^{M(i, j)}}{\sum_{k=1}^4 e^{M(i, k)}} \quad \text{for } i, j = 1, 2, 3, 4 \quad (16)$$

Soft attention is then applied over the attention matrices to compute the attentive gated representations.

$$O = P \cdot GMU_{out} \quad (17)$$

Finally, a *multiplicative gating* function is computed between the attentive gated representations.

$$GEMA_{out} = O \odot GMU_{out} \quad (18)$$

where $GEMA_{out}$ denotes the output of the Gated Emoji-aware Multimodal Attention and $\odot$ represents the element-wise matrix multiplication. Motivated by the residual skip connection

[49], $GEMA_{out}$ is concatenated with the representations ER, EIMR, EEMR, and EMR.

$$final_{rep} = \text{Concat}(GEMA_{out}, ER, EIMR, EEMR, EMR) \quad (19)$$

Finally, the concatenated representation is passed through an output layer for sarcasm prediction.

$$prediction_{sar} = \text{Softmax}(final_{rep}) \quad (20)$$

The gradients are updated based on seven losses, i.e., sarcasm ($\mathcal{L}_S$), implicit sentiment ($\mathcal{L}_{IS}$), explicit sentiment ($\mathcal{L}_{ES}$), implicit emotion ($\mathcal{L}_{IE}$), explicit emotion ($\mathcal{L}_{EE}$), emoji's sentiment ($\mathcal{L}_{E_S^m}$), and emoji's emotion ($\mathcal{L}_{E_E^m}$).

$$\begin{aligned} \mathcal{L}_{S^r} &= - \sum_{i=1}^M y_{S^r}^i \log(p_{S^r}^i) \\ \mathcal{L}_{IS} &= - \sum_{i=1}^M y_{IS}^i \log(p_{IS}^i) \\ \mathcal{L}_{ES} &= - \sum_{i=1}^M y_{ES}^i \log(p_{ES}^i) \\ \mathcal{L}_{IE} &= -(y_{IE}^i \log(p_{IE}^i) + (1 - y_{IE}^i) \log(1 - p_{IE}^i)) \\ \mathcal{L}_{EE} &= -(y_{EE}^i \log(p_{EE}^i) + (1 - y_{EE}^i) \log(1 - p_{EE}^i)) \\ \mathcal{L}_{E_S^m} &= - \sum_{i=1}^M y_{E_S^m}^i \log(p_{E_S^m}^i) \\ \mathcal{L}_{E_E^m} &= - \sum_{i=1}^M y_{E_E^m}^i \log(p_{E_E^m}^i) \end{aligned} \quad (21)$$

where p_l^i is the i th predicted output and y_l^i is the corresponding target value and $l \in [S^r, IS, ES, IE, EE, E_S^m, E_E^m]$.

6. Dataset

The original MUSTARD [11] dataset consists of conversational audio-visual utterances (total length of 3.68 h). The samples were gathered from four famous TV shows viz., Buddies, The Big Bang Theory, The Golden Girls, and Sarcasmaholics Anonymous and annotated manually. For both sarcastic/non-sarcastic labels, the dataset is balanced. This dataset has 690 samples, and each sample consists of its sarcasm label and its context (multiple utterances). Please note that the utterance in each sample (dialogue) consists of a single utterance, while its context consists of multiple utterances.

Chauhan et al. [1] manually re-annotated the MUSTARD dataset [11] to introduce sentiment and emotion labels in addition to sarcasm, named SE-MUSTARD. The authors defined two kinds of emotion and sentiment values viz., implicit and explicit. They annotated *implicit sentiment* of an utterance with the help of context. Whereas *explicit sentiment* was determined directly without any help of context. For both implicit and explicit types, they considered three sentiment classes, namely, *positive*, *negative* and *neutral* and nine emotion values, viz. anger (An), excited (Ex), fear (Fr), sad (Sd), surprised (Sp), frustrated (Fs), happy (Hp), neutral (Neu), and disgust (Dg).

We have further annotated the MUSTARD [1] dataset with extra information in the form of emojis, emoji's sentiment, and emoji's emotion. We use 25 different and most frequently used emojis on social media, which represent different emotion as well as the sentiment. We annotate three sentiment values, viz. *positive*, *negative* and *neutral*, and nine emotion values, viz. anger,

Table 2
Samples from *SEEmoji* MUSTARD.

No.	Utterances	Emoji	Sentiment	Emotion
1	<i>It's just a privilege to watch your mind at work.</i>	🤔	Negative	Disgust
2	<i>To feed the cat Rose.</i>	😸	Positive	Happy
3	<i>You're kidding, right?</i>	😬	Positive	Surprised
4	<i>Could you be more specific?</i>	😏	Neutral	Neutral
5	<i>Oh my God I love that! - Really? - NO!</i>	😡	Negative	Anger
Emojis used for annotation				
				

Table 3
Different version of MUSTARD datasets.

[illegible]

excited, fear, sad, surprised, frustrated, happy, neutral, and disgust.⁶ We show some samples from the dataset and emojis in Table 2. Also, we show all versions of MUSTARD dataset in Table 3.

As we know that sentiment and emotion of multimodal data help in better sarcasm detection [1]. To complement the sentiment and emotion information, which plays a significant role in sarcasm detection, of multimodal data, we also use emoji’s sentiment and emotion information. The idea is to capture the relation between sentiment & emotion of emojis and multimodal data and the combined effect they have in a better understanding of sarcasm.

Please note that an emoji might carry a different meaning, but the sentiment will always be the same. For example, if an emoji has negative sentiment, then different meanings will also be negative and can never be positive. So, the different meanings of any emoji can never affect sarcasm.

6.1. Annotation details

We employ four graduate students who are highly proficient in English with prior experience in labeling *emoji*, *sentiment*, *emotion*, and *sarcasm*. The guidelines for annotation, along with some examples (c.f. Table 2), were explained to the annotators before starting the annotation process. We have given the utterance and its’ context with video frames to annotators without sarcasm labels and asked them to annotate every utterance with its’ context with one emoji and corresponding sentiment and emotion⁷ of that emoji. They watched the video and annotated it accordingly. The purpose of not giving sarcasm labels is to avoid bias toward sarcasm labels.

We use 25 different and most frequently used emojis on social media and we divided these emojis into nine emotion classes. As we can see, there are three emojis per emotion approximately. So, at the time of annotation, different annotators may take different emojis from the same cluster. For example, we have three emojis for sad emotion, i.e., 😞, 😓, 😞 and if annotators use different emoji for an utterance then we randomly choose among them, because all emojis belong to sad class and has negative sentiment. If annotators take different emojis from the different clusters then we take emoji based on the majority voting scheme. We achieve an overall Fleiss' [50] kappa score of 0.82, which is considered reliable.

⁶ We shall make the datasets and codes available.

⁷ Only one emotion per utterance.

6.2. Description of emojis

We use 25 frequently used emojis on social media, and the detailed description of emojis are as follows;

6.2.1. Anger

😡: A yellow face with a frowning mouth and eyes and eyebrows scrunched downward in anger.

☹️: A red face with an angry expression; frowning mouth with eyes and eyebrows scrunched downward. It conveys more intense degrees of anger, e.g., hate or rage, than 😡 angry Face.

6.2.2. Excited

✨: The glittering flashes of sparkles. It is commonly used to indicate various positive sentiments, including love, happiness, beauty, gratitude, and excitement.

🤗: A yellow smiling face with open hands as if giving a hug. It may be used to offer thanks, support, and show love and care or to express warm, positive feelings more generally.

6.2.3. Fear

😬: A face with small, open eyes, an open frown, raised eyebrows, and a pale blue forehead, as if experiencing a cold flash.

👤: A yellow face screaming in fear, depicted by wide, white eyes, a long, open mouth, hands pressed on cheeks, and a pale blue forehead as if it has lost its color. Its expression evokes Edvard Munch's iconic painting *The Scream*.

6.2.4. Sad

😞: A yellow face with raised eyebrows and a slight frown, shedding a single, blue tear from one eye down its cheek. It may convey a moderate degree of sadness or pain.

☹️: A pensive, remorseful face saddened by life—quietly considering where things all went wrong. It is depicted as a yellow face with sad, closed eyes, furrowed eyebrows, and a slight, flat mouth.

😭: A yellow face with an open mouth wailing and streams of heavy tears flowing from closed eyes. It may convey inconsolable grief and other intense feelings, such as uncontrollable laughter, pride, or overwhelming joy.

6.2.5. Surprised

🤨: A yellow face with small, open eyes, raised eyebrows, and a small, open mouth. This emoji's meaning varies widely, but its expression is commonly taken as a surprise, embarrassment, or mild excitement.

😲: A yellow face with small, open eyes and a large, round mouth, slack with surprise or shock, as if saying Wow! or Oh my! May convey such feelings as awe or disbelief, often milder or more ironic in tone than 😱 face screaming in fear.

6.2.6. Frustrated

👊: A hand is shown pressing against the head of a person, commonly written as facepalm. It is used to display frustration or embarrassment at the ineptitude of a person or situation.

😠: A yellow face with closed eyes, furrowed eyebrows, a broad frown, and two puffs of steam blowing out of its nose as if in a huff or fuming. It may convey various negative emotions, including irritation, anger, and contempt.

6.2.7. Happy

😂: A yellow face with a big grin and scrunched, X-shaped eyes tilted on its side as if rolling on the floor laughing. It often conveys hysterical laughter more intense than 😊 face with tears of joy.

😘: A yellow winking face with puckered lips blowing a kiss is depicted as a small, red heart. It may represent a kiss goodbye or good night and generally convey feelings of love and affection.

😊: A yellow face with smiling eyes and a broad, closed smile turning up to rosy cheeks. It often expresses genuine happiness and warm, positive feelings.

😄: A yellow face with a big grin, uplifted eyebrows, and smiling eyes, each shedding a tear from laughing so hard.

❤️: A classic red love heart emoji is used to express love and romance. A similar emoji exists for the heart suit in a deck of playing cards.

😍: A yellow face with an open smile, sometimes showing teeth, and red, cartoon-styled hearts for eyes. It often conveys enthusiastic feelings of love, infatuation, and adoration, e.g., I love/am in love with this person or thing.

😁: A yellow face with smiling eyes and a full-toothed grin, as if saying Cheese! for the camera. Teeth may be smoothed-over or crosshatched. Often expresses a radiant, gratified happiness.

6.2.8. Neutral

😐: A yellow face with simple, open eyes and a flat, closed mouth. Intended to depict a neutral sentiment but is often used to convey mild irritation and concern or a deadpan sense of humor.

😑: A yellow face with flat, closed eyes and mouth. It may convey a sense of frustration or annoyance more intense than suggested by 😐 neutral face.

6.2.9. Disgust

🤢: A sickly-green face with concerned eyes and puffed, often red cheeks, as if holding back vomit. It may represent physical illness or general disgust.

🤮: A yellow face with scrunched, X-shaped eyes are spewing bright-green vomit. It may represent physical illness or disgust more intensely than 🤢 nauseated face.

💩: A swirl of brown poop, shaped like soft-serve ice cream with large, excited eyes and a big, friendly smile. It may represent feces and other bathroom topics and stand in for their many related slang terms.

7. Experiments and analysis

In this section, we describe our experimental setups and results analysis which are as follows;

7.1. Experimental setup

We implement⁸ our proposed model on the Keras and PyTorch libraries. As the evaluation metric, we employ precision (P), recall (R), and F1-score (F1) for implicit sentiment/emotion,

⁸ Please note that, we use GPUs (GPU: 1080Ti with 32 GB, RAM: 256 GB) for all experiments. Our model only takes approx 1.5 GB of GPU memory. It takes 2–3 s per epoch approximately.

explicit sentiment/emotion, emoji's sentiment, emoji's emotion, and sarcasm detection. We use *Adam* as an optimizer, *Softmax* as a classifier for implicit/explicit sentiment, emoji's sentiment, emoji's emotion, and sarcasm detection, and the *categorical cross-entropy* as a loss function. For implicit/explicit emotion recognition, we use *Sigmoid* as an activation function and optimize the *binary cross-entropy* as the loss function.

We evaluate our proposed model on the *SEEmoji MUSTARD*. We perform *grid search* to obtain the optimal hyper-parameters (c.f. Table 4). Though we aim to use a generic hyper-parameter configuration for all our experiments.

Table 4

Model configurations.

Parameters	Speaker dependent	Speaker independent
Bi-GRU	2 × 50 N, D=0.5	2 × 200 N, D=0.7
Dense layer	200 N, D=0.3	100 N, D=0.3
Activations	ReLU	
Optimizer	Adam (lr=0.001)	
Outputs	Softmax ($I_s, E_s, E_s^m, E_e^m, S^r$) & Sigmoid (I_e, E_e)	
Loss	Categorical cross-entropy ($\mathcal{L}_{S^r}, \mathcal{L}_{I_s}, \mathcal{L}_{E_s}, \mathcal{L}_{E_s^m}, \mathcal{L}_{E_e^m}$)	
	Binary cross-entropy ($\mathcal{L}_{I_e}, \mathcal{L}_{E_e}$)	
Batch	32	
Epochs	200	

We divide the whole process into four cases: (i) utterance w/o context and w/o speaker, (ii) utterance w/ context and w/o speaker, (iii) utterance w/o context and w/ speaker, and (iv) utterance w/ context and w/ speaker. Please note that, for each case, we only modify Eq. (5), and the rest model is the same.

7.1.1. Utterance w/o context and w/o speaker

We only take utterance without context and speaker information. Thus, the new Eq. (5) is as follows;

$$T_{cat} = \text{Concat}([T_u], [])$$

$$V_{cat} = \text{Concat}([V_u], [])$$

$$A_{cat} = \text{Concat}([A_u], [])$$

$$E_{cat}^m = \text{Concat}([E_u^m], [])$$

7.1.2. Utterance w/ context and w/o speaker

We take utterance with context but without speaker information. Thus, the new Eq. (5) is as follows;

$$T_{cat} = \text{Concat}([T_u], [T_c])$$

$$V_{cat} = \text{Concat}([V_u], [V_c])$$

$$A_{cat} = \text{Concat}([A_u], [])$$

$$E_{cat}^m = \text{Concat}([E_u^m], [E_c^m])$$

7.1.3. Utterance w/o context and w/ speaker

We take utterance without context but with speaker information. Thus, the new Eq. (5) is as follows;

$$T_{cat} = \text{Concat}([T_u + SP_u], [])$$

$$V_{cat} = \text{Concat}([V_u + SP_u], [])$$

$$A_{cat} = \text{Concat}([A_u + SP_u], [])$$

$$E_{cat}^m = \text{Concat}([E_u^m], [])$$

7.1.4. Utterance w/ context and w/ speaker

We take utterance with context and speaker information. Thus, the new Eq. (5) is as follows;

$$T_{cat} = \text{Concat}([T_u + SP_u], [T_c + SP_c])$$

$$V_{cat} = \text{Concat}([V_u + SP_u], [V_c])$$

$$A_{cat} = \text{Concat}([A_u + SP_u], [])$$

$$E_{cat}^m = \text{Concat}([E_u^m], [E_c^m])$$

For a fair comparison with the state-of-the-art systems [1,11], we perform the detailed experiments (i.e., unimodal, bimodal,

and trimodal) for case i, while for case ii, case iii, and case iv, we perform experiments only with the trimodal setting. Our experimental setup is mainly divided into two main parts similar to [1,11];

1. **Speaker Dependent Setup:** In this setup, a five-fold cross-validation was performed for the experiments, where each fold takes samples randomly in a stratified manner from all the TV shows.
2. **Speaker Independent Setup:** In this experiment, samples from three TV shows (i.e., The Golden Girls, Big Bang Theory, and Sarcasmaholics Anonymous) were taken in the training set. In contrast, samples from the fourth TV show (i.e., Friends) were taken in the test set. Following this step, we were able to reduce the effect of the speaker in the model.

7.2. Results and analysis

We evaluate our proposed architecture (for case i) with all the possible input combinations (c.f. Table 5), i.e., unimodal (T, A, V), bimodal (T+V, T+A, A+V) and trimodal (T+V+A). For speaker dependent setup, our proposed multitask framework (for trimodal) achieves the best precision of 77.9% (8.8⁹ points ↑ and 7.3¹⁰ points ↑), recall of 76.9% (8.9 points ↑ and 6.6 points ↑) and F1-score of 76.7% (9.3 points ↑ and 6.5 points ↑). We observe that trimodal performs better than the unimodal and bimodal.

Table 5
Experimental results for w/o context and w/o speaker setup.

Labels	Speaker dependent			Speaker independent		
	P	R	F1	P	R	F1
T	69.9	69.7	69.6	63.1	61.2	62.0
A	69.1	68.0	67.4	67.2	67.4	67.3
V	75.1	74.2	74.0	65.4	65.7	65.5
T+V	75.1	74.8	74.7	66.2	66.6	66.2
T+A	70.6	70.3	70.1	69.5	66.0	65.9
A+V	76.1	75.7	75.6	68.9	69.1	68.2
T+V+A	77.9	76.9	76.7	70.0	69.7	69.8

Similarly, for the Speaker Independent setup, we obtain precision of 70.0% (6.9¹¹ points ↑ and 0.5¹² points ↑), recall of 69.7% (8.5 points ↑ and 3.7 points ↑) and F1-score of 69.8% (7.8 points ↑ and 3.9 points ↑) for the proposed multitask model.

We further evaluate our proposed model by incorporating context and speaker information to form the three combinations of experiments viz. *utterance w/ context and w/o speaker information*, *utterance w/o context and w/ speaker information*, *utterance w/ context and w/ speaker information* (c.f. Table 6).

Table 6
Experimental results for different combination of Context-Speaker Experiments.

Setups		Speaker dependent			Speaker independent		
Context	Speaker	P	R	F1	P	R	F1
×	×	77.9	76.9	76.7	70.0	69.7	69.8
×	✓	79.9	78.8	78.7	80.7	80.1	80.0
✓	×	74.2	73.3	73.3	69.5	69.7	69.1
✓	✓	77.4	76.7	76.5	73.7	73.9	73.7

The maximum improvement of 2.0 points in precision, 1.9 points in recall, and 2.0 points in F1-score is observed when the speaker information alone is incorporated in the trimodal setup. In Speaker Independent Setup, incorporating both context and speaker information significantly improves the performance with an improvement of 10.7 points in precision, 10.4 points in

recall, and 10.2 points in F1-score. The experiments *utterance w/o context and w/o speaker information* are same as the proposed trimodal setup in Table 5.

7.3. Comparative analysis

We compare under similar experimental setups against the existing models called Baseline-1 [11], and Baseline-2 [1], which also made use of the same dataset.

7.3.1. Utterance w/o context and w/o speaker

The comparative analysis is shown in Table 7. For speaker dependent setup, our proposed multimodal multitask framework (for the trimodal setup) achieves the best precision of 77.9% (6¹³ points ↑ and 4.5¹⁴ points ↑), recall of 76.9% (5.5 points ↑ and 4.2 points ↑) and F1-score of 76.7% (5.2 points ↑ and 4.1 points ↑) for the proposed multitask model as compared to precision of 71.9%, recall of 71.4%, F1-score of 71.5% of Baseline-1 and precision of 73.4%, recall of 72.7%, F1-score of 72.6% of Baseline-2. We also observe the similar trend of performance improvement for unimodal and bimodal systems over Baseline-1.¹⁵

Similarly, for the Speaker Independent setup, we obtain precision of 70.0% (5.7 points ↑ and 0.5 points ↑), recall of 69.7% (7.1 points ↑ and 3.7 points ↑) and F1-score of 69.8% (7.0 points ↑ and 3.9 points ↑) for the proposed multitask model as compared to precision of 64.3%, recall of 62.6%, F1-score of 62.8% of Baseline-1 and precision of 69.5%, recall of 66.0%, F1-score of 65.9% of Baseline-2. Thus, we observe that emoji helps improve the performance of sarcasm detection.

7.3.2. Utterance w/o context and w/ speaker

For speaker-dependent setup, our proposed multitask framework (for trimodal setup) achieves an improvement of 2.8 points in precision, 2.1 points in recall, and 2.1 points in F1-score over Baseline-2. Similarly, for speaker-independent setup, our proposed model shows an improvement of 6.0 points in precision, 5.7 points in recall, and 5.5 points in F1-score. The results are shown in the Table 8.

7.3.3. Utterance w/ context and w/o speaker

For speaker-dependent setup (c.f. Table 8), our proposed multitask framework (for trimodal setup) demonstrates an improvement of 1.9 points in precision, 1.4 points in recall, and 1.6 points in F1-score over Baseline-2. But, for the speaker-independent setup, no improvement is observed. Hence, it can be concluded that context is not helpful under the speaker-independent setup.

7.3.4. Utterance w/ context and w/ speaker

For speaker-dependent setup (c.f. Table 8), our proposed multitask framework (for trimodal setup) shows an improvement of 1.3 points in precision, 0.9 points in recall, and 0.8 points in F1-score over Baseline-2. But, for independent speaker setup, similar to Utterance w/ context and w/o speaker, we observe no improvement in the performance. Empirical results, therefore, show that context is not helpful in speaker-independent setup, both w/ and w/o speaker information.

⁹ maximum improvement over unimodal (A).

¹⁰ maximum improvement over bimodal (T+A).

¹¹ maximum improvement over unimodal (T).

¹² maximum improvement over bimodal (T+A).

¹³ w.r.t. Baseline-1.

¹⁴ w.r.t. Baseline-2.

¹⁵ The results of unimodal system are not available for Baseline-2.

Table 7Comparative analysis between our proposed model, Baseline-1 and Baseline-2 for the *w/o context* and *w/o speaker* setup.

Labels	Speaker dependent									Speaker independent								
	Proposed			Baseline-1 [11]			Baseline-2 [1]			Proposed			Baseline-1 [11]			Baseline-2 [1]		
	P	R	F1	P	R	F1	P	R	F1	P	R	F1	P	R	F1	P	R	F1
T	69.9	69.7	69.6	65.1	64.6	64.6	–	–	–	63.1	61.2	62.0	60.9	59.6	59.8	–	–	–
A	69.1	68.0	67.4	65.9	64.6	64.6	–	–	–	67.2	67.4	67.3	65.1	62.6	62.7	–	–	–
V	75.1	74.2	74.0	68.1	67.4	67.4	–	–	–	65.4	65.7	65.5	54.9	53.4	53.6	–	–	–
T+V	75.1	74.8	74.7	72.0	71.6	71.6	72.7	71.9	71.6	66.2	66.6	66.2	62.2	61.5	61.7	65.5	65.5	65.7
T+A	70.6	70.3	70.1	66.6	66.2	66.2	62.2	61.1	59.6	69.5	66.0	65.9	64.7	62.9	63.1	59.1	60.0	50.3
A+V	76.1	75.7	75.6	66.2	65.7	65.7	72.7	71.9	71.8	68.9	69.1	68.2	64.1	61.8	61.9	65.6	63.8	63.9
T+V+A	77.9	76.9	76.7	71.9	71.4	71.5	73.4	72.7	72.6	70.0	69.7	69.8	64.3	62.6	62.8	69.5	66.0	65.9

Table 8

Comparative analysis between our proposed model and Baseline-2 for different combination of Context-Speaker Experiments. Baseline-1 results for trimodal setting are not available.

Setups		Speaker dependent						Speaker independent					
Context	Speaker	Proposed			Baseline-2 [1]			Proposed			Baseline-2 [1]		
		P	R	F1	P	R	F1	P	R	F1	P	R	F1
x	x	77.9	76.9	76.7	73.4	72.7	72.6	70.0	69.7	69.8	69.5	66.0	65.9
x	✓	79.9	78.8	78.7	77.1	76.7	76.6	80.7	80.1	80.0	74.7	74.4	74.5
✓	x	74.2	73.3	73.3	72.3	71.9	71.7	69.5	80.1	69.1	71.5	71.3	70.5
✓	✓	77.4	76.7	76.5	76.1	75.8	75.7	73.7	73.9	73.7	74.9	75.0	74.7

Table 9

Comparison between samples from the dataset, w/ Emoji and w/o Emoji.

	Utterances	W/o Emoji		W/ Emoji	
		S^r		E^m	S^r
1	Oh, I'm so glad you asked it like that. You.	Non-sarcastic		🤔	Sarcastic
2	We can? Ok I am trying that.	Non-sarcastic		😏	Sarcastic
3	Wow you look just like your son, Mrs. Tribbiani	Non-sarcastic		😬	Sarcastic
4	You got off to a really good start with the group.	Non-sarcastic		😏	Sarcastic
5	Maybe it's a contest, you know? Like "Collect all five".	Non-sarcastic		😏	Sarcastic

7.4. Ablation study

To understand the effect of emoji and proposed GEMA, we perform an ablation study for our proposed model. The results are shown in Table 10. For speaker-dependent setup, our proposed model shows an improvement of 5.7 points in precision, 5.2 points in recall, and 5.1 points in F1-score over the model w/o emojis. We also observe an improvement of 3.0 points in precision, 1.6 points in recall, and 1.6 points in F1-score over the model w/o GMU module. Similarly, We observe an improvement of 3.5 points in precision, 2.0 points in recall, and 2.0 points in F1-score over the model w/o GEMA module.

Table 10

Ablation study.

Setup	Speaker dependent			Speaker independent		
	P	R	F1	P	R	F1
W/o Emoji	72.2	71.7	71.6	64.8	65.1	64.8
W/o GEMA	74.4	74.9	74.7	67.7	67.9	67.8
W/o GMU	74.9	75.3	75.1	68.2	68.4	68.3
Proposed	77.9	76.9	76.7	70.0	69.7	69.8

While for speaker-independent setup, our proposed model shows an improvement of 5.3 points in precision, 4.6 points in recall, and 5.0 points in F1-score over the model w/o emojis. We also observe an improvement of 1.8 points in precision, 1.3 points in recall, and 1.5 points in F1-score over the model w/o GMU module. Similarly, We also observe an improvement of 2.4 points in precision, 1.8 points in recall, and 2.0 points in F1-score over the model w/o GEMA module.

7.5. Impact of emoji

Empirically, we have shown that emoji helps sarcasm (C.f. Table 10). We take some examples from the dataset (c.f. Table 9), which are sarcastic, to show the effect of emojis. Each example has positive implicit/explicit sentiment. The predictions made by the model, proposed w/o emoji, are incorrect for the sarcasm but correct for the implicit and explicit sentiment.

When emojis are used, the model correctly predicts all the examples as sarcastic. We observe that emoji's sentiment plays an important role in sarcasm detection. The sentiment displayed by the emojis is negative. This helps the model to understand the contrast between the sentiments displayed by the utterance and the emoji. Thus, it correctly interprets that the utterance is sarcastic.

7.6. Impact of GEMA

Empirically, we have shown the effectiveness of the GEMA (C.f. Table 10). We show the heatmap of an utterance "Oh, I'm so glad you asked it like that. You". (c.f. Table 9), and we have already shown that emoji 🤔 helps to predict this utterance as sarcastic. To prove this, we show the attention heatmap for this utterance in Fig. 4. We see that ER contributes more than others which means emoji is more evident for this utterance to predict correctly. Thus, this also proves our hypothesis that emoji help sarcasm.

7.7. Error analysis

In this section, we perform error analysis for our proposed model. We take some samples incorrectly specified by our proposed model and analyze the model's shortcomings.

Table 11
Error analysis: some incorrectly predicted samples by the proposed model.

	Utterances	Actual	Predicted	
		S^r	E^m	S^r
1	<i>A tshirt that says - I don't belong here.</i>	Sarcastic	😐	Non-sarcastic
2	<i>Yes you can. You're thinking about time, you can't go back in time.</i>	Non-sarcastic	😐	Sarcastic
3	<i>I thought if I littered, that crying Indian might come by and save us.</i>	Sarcastic	😐	Non-sarcastic

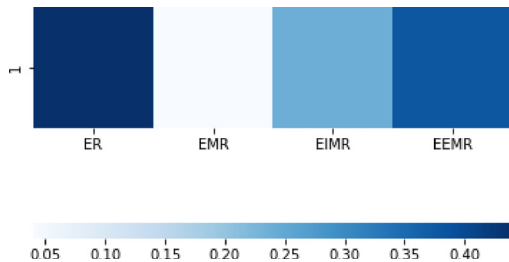


Fig. 4. The heatmap represents attention weights of a particular utterance across ER, EMR, EIMR and EEMR.

We first consider the first example (c.f. Table 11). It has neutral implicit/explicit sentiment and emotion. The accompanying emoji is 😐 (expressionless) which also happens to have neutral sentiment and emotion. The model does not see any of this information as a reason to predict the example as sarcastic.

For the second and third examples, the implicit/explicit sentiment of the examples is positive, and the emoji is 😐 (expressionless). Even though the information for sentiment and emojis are the same for both, the second example is non-sarcastic, while the third example is sarcastic. With this, the model fails to learn the subtle difference between the examples as the emojis do not provide any additional distinguishable information to the model about the examples during training.

8. Conclusion

In this paper, we have created *SEEmoji MUSTARD* by manually annotating an existing MUSTARD dataset with emoji, emoji's sentiment, and emotion labels. In our multitask framework, sarcasm is treated as the primary task, whereas emotion and sentiment analysis are considered secondary tasks. We have proposed an effective deep learning-based emoji-aware-multitask learning framework for sarcasm prediction. We have also proposed a Gated Emoji-aware Multimodal Attention mechanism for Sarcasm prediction. Empirical results of our proposed model on *SEEmoji MUSTARD* achieve state-of-the-art performance over the existing methods.

During the annotation, we found that the dataset is very small for a complex architecture to learn a complex problem like sarcasm. We think that increasing the size of the dataset by annotating more samples should be helpful to gain improvement in performance.

9. Ethical declaration

The dataset *SEEmoji MUSTARD* is freely available and will be used only for the purpose of academic research. We create *SEEmoji MUSTARD* by extending the MUSTARD dataset by annotating Emoji, Emoji's sentiment, and Emoji's emotion. The annotation for extending the dataset was done by human experts, who are the regular employee of our research group. There are no other issues to declare.

CRedit authorship contribution statement

Dushyant Singh Chauhan: Conceptualization, Methodology, Data curation, Writing – original draft, Supervision, Validation. **Gopendra Vikram Singh:** Data curation, Writing – review & editing. **Aseem Arora:** Data curation, Writing – review & editing. **Asif Ekbal:** Writing – review & editing, Investigation. **Pushpak Bhattacharyya:** Writing – review & editing, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

Dushyant Singh Chauhan acknowledges the support of the Prime Minister Research Fellowship (PMRF), Govt. of India. Asif Ekbal acknowledges the Young Faculty Research Fellowship (YFRF), supported by Visvesvaraya Ph.D. Scheme for Electronics and IT, Ministry of Electronics and Information Technology (Meit/8Y), Government of India, being implemented by Digital India Corporation (formerly Media Lab Asia).

References

- [1] D.S. Chauhan, D. S. R., A. Ekbal, P. Bhattacharyya, Sentiment and emotion help sarcasm? A multi-task learning framework for multi-modal sarcasm, sentiment and emotion analysis, in: Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics, Association for Computational Linguistics, Online, 2020, pp. 4351–4360, <http://dx.doi.org/10.18653/v1/2020.acl-main.401>, URL <https://www.aclweb.org/anthology/2020.acl-main.401>.
- [2] D. Hazarika, S. Poria, S. Gorantla, E. Cambria, R. Zimmermann, R. Mihalcea, Cascade: Contextual sarcasm detection in online discussion forums, 2018, arXiv preprint [arXiv:1805.06413](https://arxiv.org/abs/1805.06413).
- [3] Y.A. Kolchinski, C. Potts, Representing social media users for sarcasm detection, 2018, arXiv preprint [arXiv:1808.08470](https://arxiv.org/abs/1808.08470).
- [4] A.K. Jena, A. Sinha, R. Agarwal, C-net: Contextual network for sarcasm detection, in: Proceedings of the Second Workshop on Figurative Language Processing, 2020, pp. 61–66.
- [5] R. Misra, P. Arora, Sarcasm detection using hybrid neural network, 2019, arXiv preprint [arXiv:1908.07414](https://arxiv.org/abs/1908.07414).
- [6] M.S. Razali, A.A. Halin, N.M. Norowi, S.C. Doraisamy, The importance of multimodality in sarcasm detection for sentiment analysis, in: 2017 IEEE 15th Student Conference on Research and Development, SCORed, IEEE, 2017, pp. 56–60.
- [7] D. Das, A Multimodal Approach to Sarcasm Detection on Social Media (Ph.D. thesis), Missouri State University, 2019.
- [8] R. Schifanella, P. de Juan, J. Tetreault, L. Cao, Detecting sarcasm in multimodal social platforms, in: Proceedings of the 24th ACM International Conference on Multimedia, ACM, 2016, pp. 1136–1145.
- [9] Y. Cai, H. Cai, X. Wan, Multi-modal sarcasm detection in Twitter with hierarchical fusion model, in: Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics, 2019, pp. 2506–2515.
- [10] X. Wang, X. Sun, T. Yang, H. Wang, Building a bridge: A method for image-text sarcasm detection without pretraining on image-text data, in: Proceedings of the First International Workshop on Natural Language Processing beyond Text, Association for Computational Linguistics, Online, 2020, pp. 19–29, <http://dx.doi.org/10.18653/v1/2020.nlpbt-1.3>, URL <https://www.aclweb.org/anthology/2020.nlpbt-1.3>.

- [11] S. Castro, D. Hazarika, V. Pérez-Rosas, R. Zimmermann, R. Mihalcea, S. Poria, Towards multimodal sarcasm detection (an ‘Obviously Perfect’ paper), 2019, arXiv preprint [arXiv:1906.01815](https://arxiv.org/abs/1906.01815).
- [12] S. Attardo, J. Eisterhold, J. Hay, I. Poggi, Multimodal markers of irony and sarcasm, *Humor* 16 (2) (2003) 243–260.
- [13] D.S. Chauhan, D. S. R. A. Ekbal, P. Bhattacharyya, All-in-one: A deep attentive multi-task learning framework for humour, sarcasm, offensive, motivation, and sentiment on memes, in: Proceedings of the 1st Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics and the 10th International Joint Conference on Natural Language Processing, Association for Computational Linguistics, Suzhou, China, 2020, pp. 281–290, URL <https://www.aclweb.org/anthology/2020.aacl-main.31>.
- [14] H. Pan, Z. Lin, P. Fu, Y. Qi, W. Wang, Modeling intra and inter-modality incongruity for multi-modal sarcasm detection, in: Findings of the Association for Computational Linguistics: EMNLP 2020, Association for Computational Linguistics, Online, 2020, pp. 1383–1392, [http://dx.doi.org/10.18653/v1/2020.findings-emnlp.124](https://dx.doi.org/10.18653/v1/2020.findings-emnlp.124), URL <https://www.aclweb.org/anthology/2020.findings-emnlp.124>.
- [15] D. Ghosal, M.S. Akhtar, D.S. Chauhan, S. Poria, A. Ekbal, P. Bhattacharyya, Contextual inter-modal attention for multi-modal sentiment analysis, in: Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing, Association for Computational Linguistics, Brussels, Belgium, 2018, pp. 3454–3466, URL <http://www.aclweb.org/anthology/D18-1382>.
- [16] S. Sangwan, D.S. Chauhan, M. Akhtar, A. Ekbal, P. Bhattacharyya, et al., Multi-task gated contextual cross-modal attention framework for sentiment and emotion analysis, in: International Conference on Neural Information Processing, Springer, 2019, pp. 662–669.
- [17] M. Soleymani, D. Garcia, B. Jou, B. Schuller, S.-F. Chang, M. Pantic, A survey of multimodal sentiment analysis, *Image Vis. Comput.* 65 (2017) 3–14.
- [18] D.S. Chauhan, M.S. Akhtar, A. Ekbal, P. Bhattacharyya, Context-aware interactive attention for multi-modal sentiment and emotion analysis, in: Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing, EMNLP-IJCNLP, Association for Computational Linguistics, Hong Kong, China, 2019, pp. 5651–5661.
- [19] D.S. Chauhan, A. Ekbal, P. Bhattacharyya, An efficient fusion mechanism for multimodal low-resource setting, in: Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval, SIGIR ’22, Association for Computing Machinery, New York, NY, USA, 2022, pp. 2583–2588, [http://dx.doi.org/10.1145/3477495.3531900](https://dx.doi.org/10.1145/3477495.3531900).
- [20] R. Kaur, S. Kautish, Multimodal sentiment analysis: A survey and comparison, *Int. J. Serv. Sci. Manage. Eng. Technol. (IJSSMET)* 10 (2) (2019) 38–58.
- [21] N. Blanchard, D. Moreira, A. Bharati, W.J. Scheirer, Getting the subtext without the text: Scalable multimodal sentiment classification from visual and acoustic modalities, 2018, arXiv preprint [arXiv:1807.01122](https://arxiv.org/abs/1807.01122).
- [22] V. Pérez-Rosas, R. Mihalcea, L.-P. Morency, Utterance-level multimodal sentiment analysis, in: Proceedings of the 51st Annual Meeting of the Association for Computational Linguistics, Volume 1: Long Papers, 2013, pp. 973–982.
- [23] S. Poria, D. Hazarika, N. Majumder, G. Naik, E. Cambria, R. Mihalcea, MELD: A multimodal multi-party dataset for emotion recognition in conversations, in: Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics, Association for Computational Linguistics, Florence, Italy, 2019, pp. 527–536, [http://dx.doi.org/10.18653/v1/P19-1050](https://dx.doi.org/10.18653/v1/P19-1050), URL <https://www.aclweb.org/anthology/P19-1050>.
- [24] M. Soleymani, M. Pantic, T. Pun, Multimodal emotion recognition in response to videos, *IEEE Trans. Affect. Comput.* 3 (2) (2011) 211–223.
- [25] M.S. Akhtar, D.S. Chauhan, D. Ghosal, S. Poria, A. Ekbal, P. Bhattacharyya, Multi-task learning for multi-modal emotion recognition and sentiment analysis, in: Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Vol. 1, Long and Short Papers, Association for Computational Linguistics, Minneapolis, Minnesota, 2019, pp. 370–379, [http://dx.doi.org/10.18653/v1/N19-1034](https://dx.doi.org/10.18653/v1/N19-1034), URL <https://www.aclweb.org/anthology/N19-1034>.
- [26] W. Dai, Z. Liu, T. Yu, P. Fung, Modality-transferable emotion embeddings for low-resource multimodal emotion recognition, in: Proceedings of the 1st Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics and the 10th International Joint Conference on Natural Language Processing, Association for Computational Linguistics, Suzhou, China, 2020, pp. 269–280, URL <https://www.aclweb.org/anthology/2020.aacl-main.30>.
- [27] J.-B. Delbrouck, N. Tits, M. Brousmiche, S. Dupont, A transformer-based joint-encoding for emotion recognition and sentiment analysis, in: Second Grand-Challenge and Workshop on Multimodal Language, Challenge-HML, Association for Computational Linguistics, Seattle, USA, 2020, pp. 1–7, [http://dx.doi.org/10.18653/v1/2020.challengehml-1.1](https://dx.doi.org/10.18653/v1/2020.challengehml-1.1), URL <https://www.aclweb.org/anthology/2020.challengehml-1.1>.
- [28] W.Y. Choi, K.Y. Song, C.W. Lee, Convolutional attention networks for multimodal emotion recognition from speech and text data, in: Proceedings of Grand Challenge and Workshop on Human Multimodal Language, Challenge-HML, Association for Computational Linguistics, Melbourne, Australia, 2018, pp. 28–34, [http://dx.doi.org/10.18653/v1/W18-3304](https://dx.doi.org/10.18653/v1/W18-3304), URL <https://www.aclweb.org/anthology/W18-3304>.
- [29] D. Hazarika, S. Poria, R. Mihalcea, E. Cambria, R. Zimmermann, Icon: Interactive conversational memory network for multimodal emotion detection, in: Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing, 2018, pp. 2594–2604.
- [30] G. Sharma, A. Dhali, A survey on automatic multimodal emotion recognition in the wild, in: Advances in Data Science: Methodologies and Applications, Springer, 2021, pp. 35–64.
- [31] A.A.M. Shueb, G. de Melo, EmoTag1200: Understanding the association between emojis and emotions, in: Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing, EMNLP, Association for Computational Linguistics, Online, 2020, pp. 8957–8967, [http://dx.doi.org/10.18653/v1/2020.emnlp-main.720](https://dx.doi.org/10.18653/v1/2020.emnlp-main.720), URL <https://www.aclweb.org/anthology/2020.emnlp-main.720>.
- [32] A. Illendula, A. Sheth, Multimodal emotion classification, in: Companion Proceedings of the 2019 World Wide Web Conference, ACM, 2019, pp. 439–449.
- [33] F. Barbieri, M. Ballesteros, H. Saggion, Are emojis predictable? 2017, arXiv preprint [arXiv:1702.07285](https://arxiv.org/abs/1702.07285).
- [34] S. Jin, T. Pedersen, Duluth urop at semeval-2018 task 2: Multilingual emoji prediction with ensemble learning and oversampling, 2018, arXiv preprint [arXiv:1805.10267](https://arxiv.org/abs/1805.10267).
- [35] S. Santhanam, V. Srinivasan, S. Glass, S. Shaikh, I stand with you: Using emojis to study solidarity in crisis events, 2019, arXiv preprint [arXiv:1907.08326](https://arxiv.org/abs/1907.08326).
- [36] G. Guibon, M. Ochs, P. Bellot, Emoji recommendation in private instant messages, in: Proceedings of the 33rd Annual ACM Symposium on Applied Computing, 2018, pp. 1821–1823.
- [37] S. Wijeratne, L. Balasuriya, A. Sheth, D. Doran, Emojinet: An open service and api for emoji sense discovery, in: Eleventh International AAAI Conference on Web and Social Media, 2017.
- [38] W. Hussien, M. Al-Ayyoub, Y. Tashtoush, M. Al-Kabi, On the use of emojis to train emotion classifiers, 2019, arXiv preprint [arXiv:1902.08906](https://arxiv.org/abs/1902.08906).
- [39] Z. Al-Halah, A. Aitken, W. Shi, J. Caballero, Smile, be happy:) emoji embedding for visual sentiment analysis, in: Proceedings of the IEEE International Conference on Computer Vision Workshops, 2019.
- [40] B. Felbo, A. Mislove, A. Søgaard, I. Rahwan, S. Lehmann, Using millions of emoji occurrences to learn any-domain representations for detecting sentiment, emotion and sarcasm, 2017, arXiv preprint [arXiv:1708.00524](https://arxiv.org/abs/1708.00524).
- [41] Y. Chen, J. Yuan, Q. You, J. Luo, Twitter sentiment analysis via bi-sense emoji embedding and attention-based LSTM, in: Proceedings of the 26th ACM International Conference on Multimedia, 2018, pp. 117–125.
- [42] X. Zhou, W.Y. Wang, Mojital: Generating emotional responses at scale, 2017, arXiv preprint [arXiv:1711.04090](https://arxiv.org/abs/1711.04090).
- [43] W. Ma, R. Liu, L. Wang, S. Vosoughi, Emoji prediction: Extensions and benchmarking, 2020, arXiv preprint [arXiv:2007.07389](https://arxiv.org/abs/2007.07389).
- [44] F. Barbieri, M. Ballesteros, F. Ronzano, H. Saggion, Multimodal emoji prediction, 2018, arXiv preprint [arXiv:1803.02392](https://arxiv.org/abs/1803.02392).
- [45] B. Eisner, T. Rocktäschel, I. Augenstein, M. Bošnjak, S. Riedel, emoji2vec: Learning emoji representations from their description, 2016, arXiv preprint [arXiv:1609.08359](https://arxiv.org/abs/1609.08359).
- [46] A. Joulin, E. Grave, P. Bojanowski, M. Douze, H. Jégou, T. Mikolov, FastText.zip: Compressing text classification models, 2016, arXiv preprint [arXiv:1612.03651](https://arxiv.org/abs/1612.03651).
- [47] K. Cho, B. van Merriënboer, D. Bahdanau, Y. Bengio, On the properties of neural machine translation: Encoder-decoder approaches, 2014, CoRR, [arXiv:1409.1259](https://arxiv.org/abs/1409.1259).
- [48] J. Arevalo, T. Solorio, M. Montes-y Gómez, F.A. González, Gated multimodal units for information fusion, 2017, arXiv preprint [arXiv:1702.01992](https://arxiv.org/abs/1702.01992).
- [49] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2016, pp. 770–778.
- [50] J.L. Fleiss, Measuring nominal scale agreement among many raters, *Psychol. Bull.* 76 (1971) 378–382.